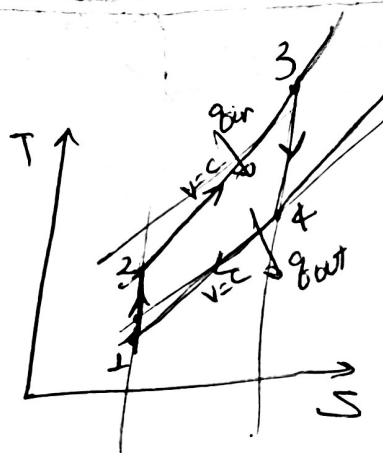
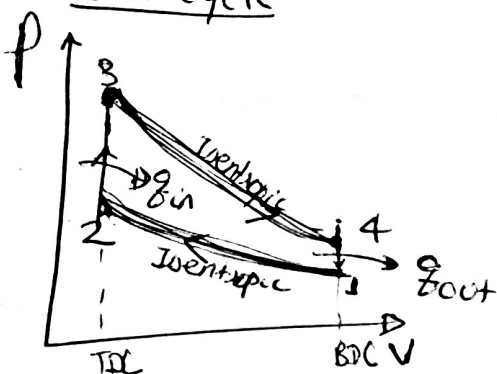


Otto cycle



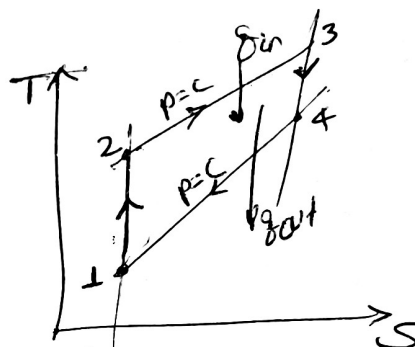
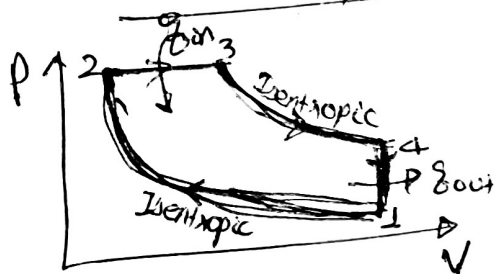
- 1-2 Isentropic compression
- 2-3 $V=C$ heat addition
- 3-4 Isentropic expansion
- 4-1 $V=C$ heat rejection

closed system
 $\Delta KE, \Delta PE \approx 0$

Energy balance

* k = specific heat ratio

Diesel cycle:



* r_c = cutoff ratio = $\frac{V_3}{V_2} = \frac{V_3}{V_1}$
 (Ratio of the cylinder volumes before & after the combustion process.)

$$\eta_{th, Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$$

$$\eta_{th, Otto} = 1 - \frac{1}{r^{k-1}}$$

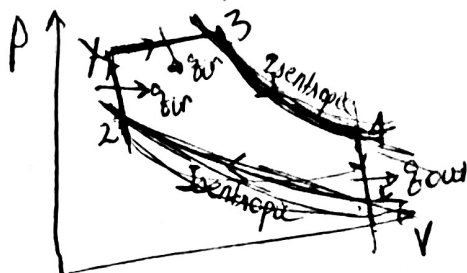
Under cold air standard assumptions

$$\eta_{th, Otto} > \eta_{th, Diesel}$$

r = same for Otto & Diesel

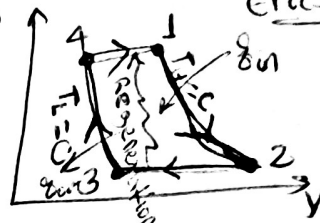
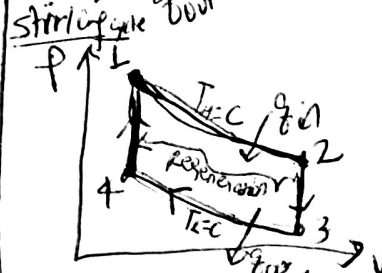
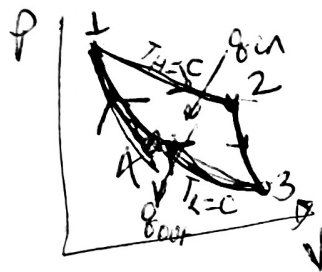
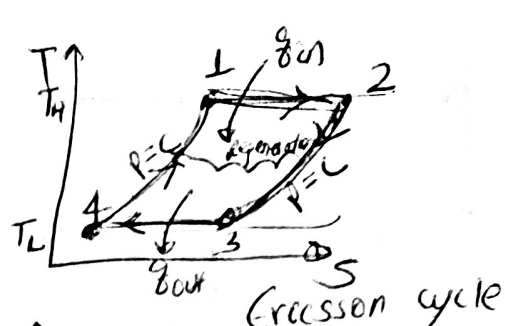
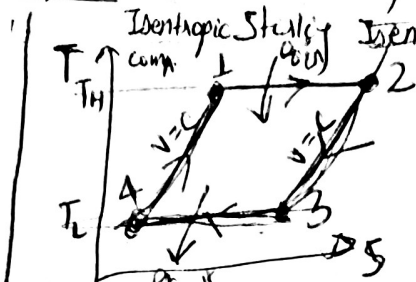
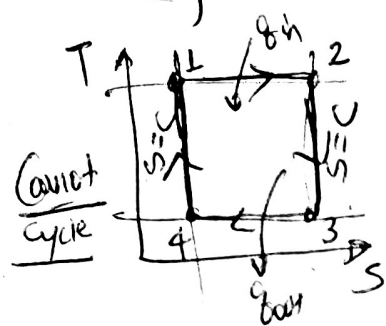
Always $P > 1$

Dual cycle



Stirling & Ericsson cycles:

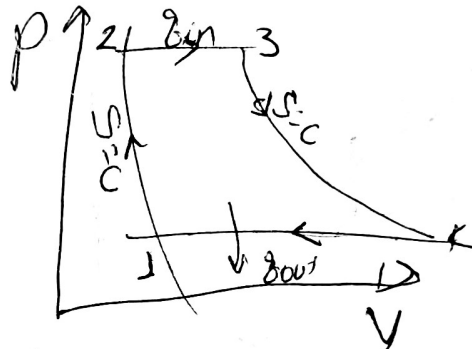
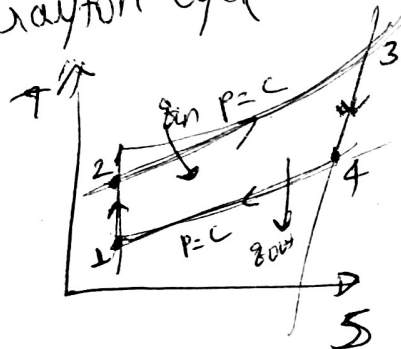
At $T = C$; heat addition & rejection



Aircraft propulsion
Electrical power generation

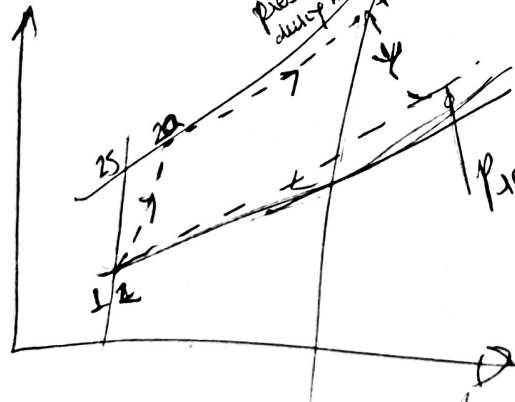
Gas turbine cycles

Brayton cycle



$r_p = \text{Pressure ratio}$
Pressure drop during heat addition

$$\left(\frac{1000}{300} \right)^{1.4} / (2(0.4))$$

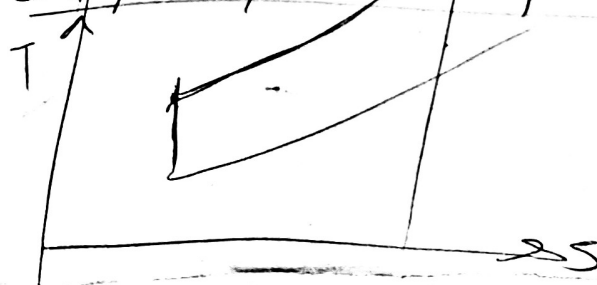


Pressure drop during heat rejection

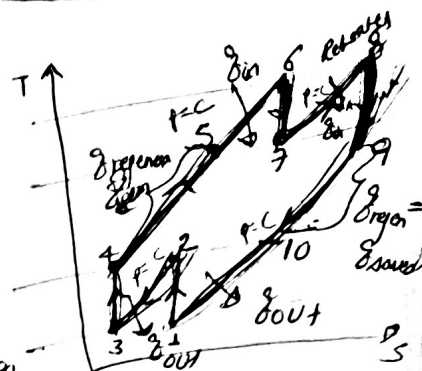
$$\eta_c = \frac{w_s}{w_a} \approx \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

$$\eta_T = \frac{w_a}{w_s} \approx \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

Brayton cycle with regeneration



- Brayton cycle

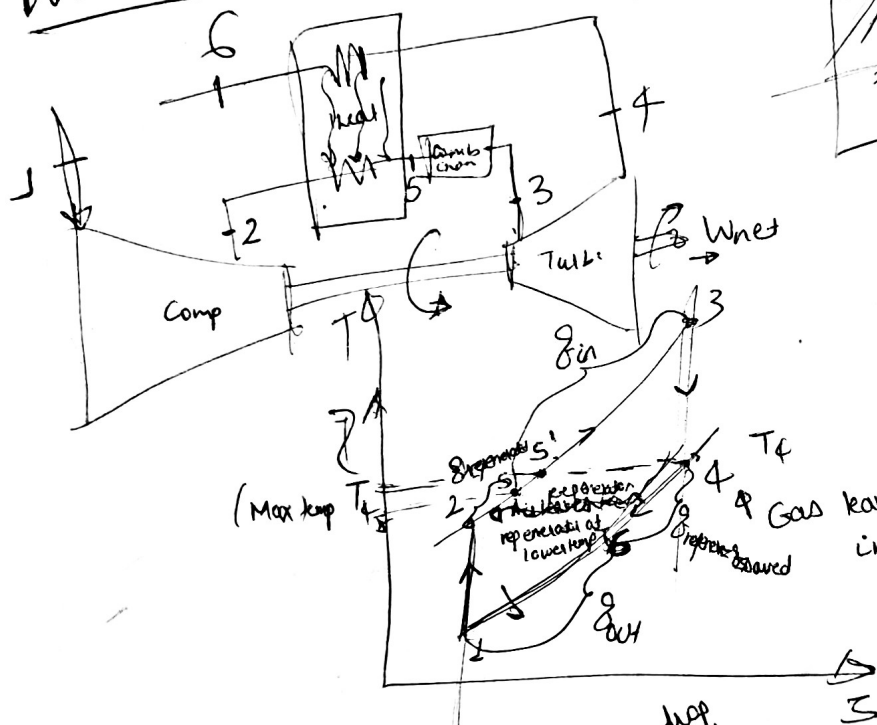


$W_{in} \uparrow$ when $W_{out} \uparrow$ → Best performance
 Equal pressure ratios are maintained across each stage

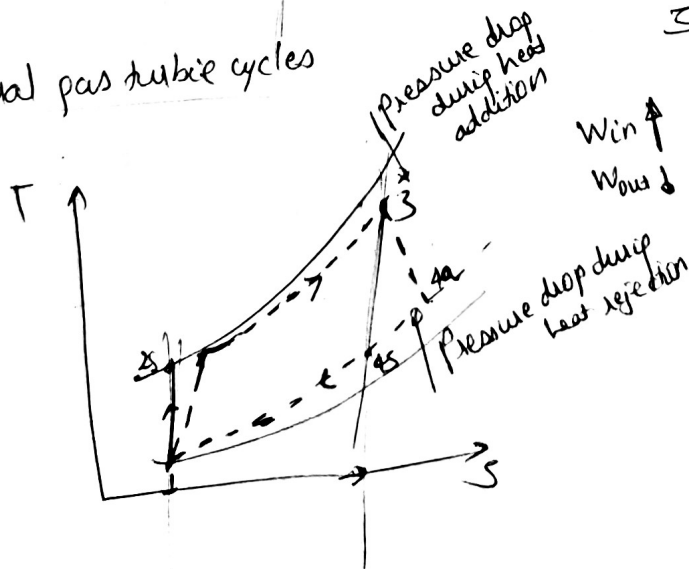
$$\frac{P_2}{P_1} = \frac{P_4}{P_3}$$

$$\frac{p_6^f}{p_7} = \frac{p_8}{h_7}$$

With Regenerator



Actual pas tube cycles



$T_4 \rightarrow$ highest temp the regenerator can have.

$T_k = T_q \Rightarrow$ Linear (ideal) case

Assuming well insulation
AKE, $A_{PE} \approx 0$

$$\delta_{\text{rejen, max}} = h_1' - h_2$$

2 refer, act

$$C = \frac{g_{regen, act}}{g_{regen, max}}$$

Ideal gas model - Constant specific heats

$$U_2 - U_1 = C_v(T_2 - T_1)$$

$$h_2 - h_1 = C_p(T_2 - T_1)$$

$$S_2 - S_1 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)$$

$$S = f(T, P)$$

$$\text{or} \\ S = f(T, V)$$

Isentropic process $s = \text{const}$

$$S_2 - S_1 = 0 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$$

$$C_p \ln\left(\frac{T_2}{T_1}\right) = R \ln\left(\frac{P_2}{P_1}\right)$$

$$\text{or } \ln\left(\frac{T_2}{T_1}\right) = \frac{R}{C_p} \ln\left(\frac{P_2}{P_1}\right)$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{R}{C_p}} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$\frac{T_2}{T_1} = \ln \left[\left(\frac{P_2}{P_1}\right)^{\frac{R}{C_p}} \right] = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$R = C_p - C_v$$

$$k = \frac{C_p}{C_v}$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^k$$

$$(S_1 = S_2, \text{ideal gas, constant } k)$$

$$(S_1 = S_2, \text{ideal gas, constant } k)$$

$$(S_1 = S_2, \text{ideal gas, constant } k)$$

Cold Air standard Assumptions: 2-continued

$$At 300K = 17^\circ C$$

Table A-2

$$C_p = 1.005 kJ/kg \cdot K$$

$$C_v = 0.718 kJ/kg \cdot K$$

Process 1-2: Isentropic compression:

$$W_{12} = U_2 - U_1$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1} = (r)^{k-1} \quad ; \text{ where } \boxed{r = \frac{V_1}{V_2} = \frac{V_4}{V_3}}$$

$$(*) T_2 = \underline{\hspace{2cm}}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^k = (r)^k$$

or Eqn of state

$$P_1 V_1 = RT_1 \rightarrow R = \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} = R$$

$$(*) P_2 = \underline{\hspace{2cm}}$$

$$P_2 V_2 = RT_2$$

$$P_2 = \frac{P_1 V_1 T_2}{T_1 V_2} = \frac{P_1}{T_1} \left(\frac{V_1}{V_2} \right) T_2$$

$$P_2 = P_1 \left(\frac{T_2}{T_1} \right) r$$

$$(*) \frac{P_3 V_3}{T_3} = \frac{P_2 V_2}{T_2}$$

$$Q_{in} = Q_{2-3} = U_3 - U_2$$

$$Q_{2-3} = C_v (T_3 - T_2)$$

$$800 kJ/kg = 0.718 kJ/kg \cdot K (T_3 - T_2)$$

$$(*) P_3 = P_2 \left(\frac{T_3}{T_2} \right)^{\frac{1}{k}}$$

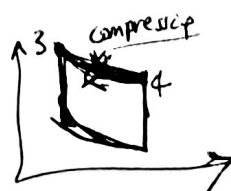
$$(*) T_3 = \underline{\hspace{2cm}}$$

Process 3-4

Expansion

So go up to 3-4 since it is not compressing.
compression ratio will be $\frac{1}{r}$

$$\frac{T_4}{T_3} = \left(\frac{1}{r} \right)^{k-1} \text{ or } \frac{T_3}{T_4} = (r)^{k-1}$$



$$P_4 V_4 = \frac{P_3 V_3}{T_3} \Rightarrow P_4 = P_3 \left(\frac{T_4}{T_3} \right)^{\frac{1}{k}} \left(\frac{V_3}{V_4} \right)^{\frac{1}{k}}$$

$$1 \rightarrow 2 : U_2 - U_1 = W_{12} = C_v (T_2 - T_1)$$

$$2 \rightarrow 3 : U_3 - U_2 = Q_{23}$$

$$C_v (T_3 - T_2) = Q_{23}$$

$$3 \rightarrow 4 : U_4 - U_3 = W_{34} = C_v (T_4 - T_3)$$

$$(*) W_{net} = W_{12} + W_{23} + W_{34} + W_{41}$$

$$= (-ve) + 0 + (+ve) + 0$$

$$= -C_v (T_2 - T_1) + W_{3-4}$$

$$(*) \eta_{th} = \frac{W_{net}}{Q_{23}} = \underline{\hspace{2cm}} \%$$

$$\eta_{th, Otto} = 1 - \frac{1}{r^{k-1}} = 1 - \frac{1}{(1.8)^{1.4}} = 56.5\% = 0.565$$

Polytropic process

$$(q_{in} - q_{out}) + \left(\frac{W_{out} - W_{in}}{T} \right) = \Delta U$$

$$\rightarrow = 0 \text{ Isentropic}$$

$$\rightarrow \neq 0 \text{ Polytropic } PV^n = C$$

$$W = \int_{in}^{final} P dV = \int_{in}^{final} \frac{C}{V^n} dV \quad \left| P = \frac{C}{V^n} \right.$$

$$= C \int_i^f \frac{1}{V^n} dV$$

$$= \frac{1}{n-1} (P_f V_f - P_i V_i)$$

$$\left| \begin{aligned} PV &= RT \\ P &= \frac{RT}{V} \end{aligned} \right.$$

$$= \frac{1}{n-1} \left(\frac{RT_f V_f}{V_f} - \frac{RT_i V_i}{V_i} \right)$$

$$W = \frac{R}{n-1} (T_f - T_i)$$

$$W_{in} = W_{12} = \frac{R}{n-1} (T_2 - T_1)$$

$$W_{out} = W_{34} = \frac{R}{n-1} (T_4 - T_3)$$

$$W_{net} = W_{out} - W_{in}$$

Isentropic: $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

Polytropic $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{n-1}$

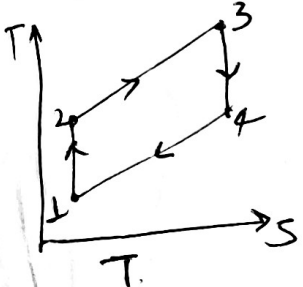
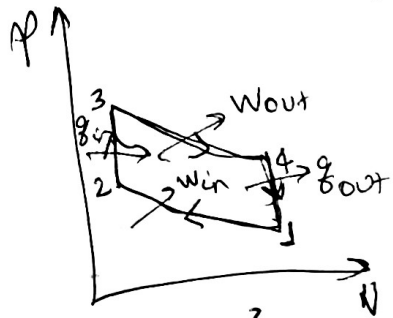
$$T_2 = T_1 (r)^{n-1} : T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$$

$$\eta = \frac{W_{net}}{q_{in}} \rightarrow q_{2-3}$$

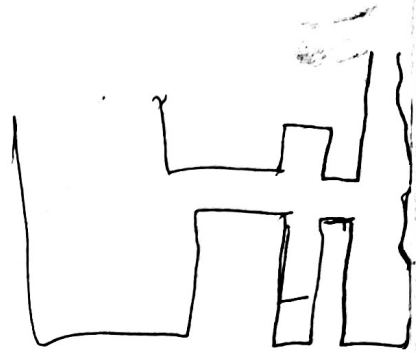
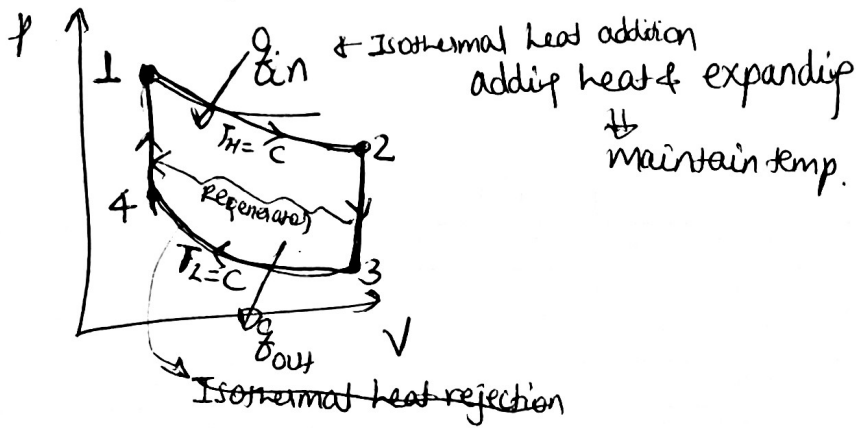
$$\eta_{MEP} = \frac{W_{net}}{V_1 - V_2}$$

State 1
 $P_1 =$
 $T_1 =$
 $V_1 = \frac{RT_1}{P_1} =$

State 2
 $T_2 = T_1 (r)^{n-1}$
 $T_4 = T_3 \left(\frac{1}{r} \right)^{n-1}$



Stirling Engine $\rightarrow 2 \rightarrow \eta_{\text{Carnot}}$

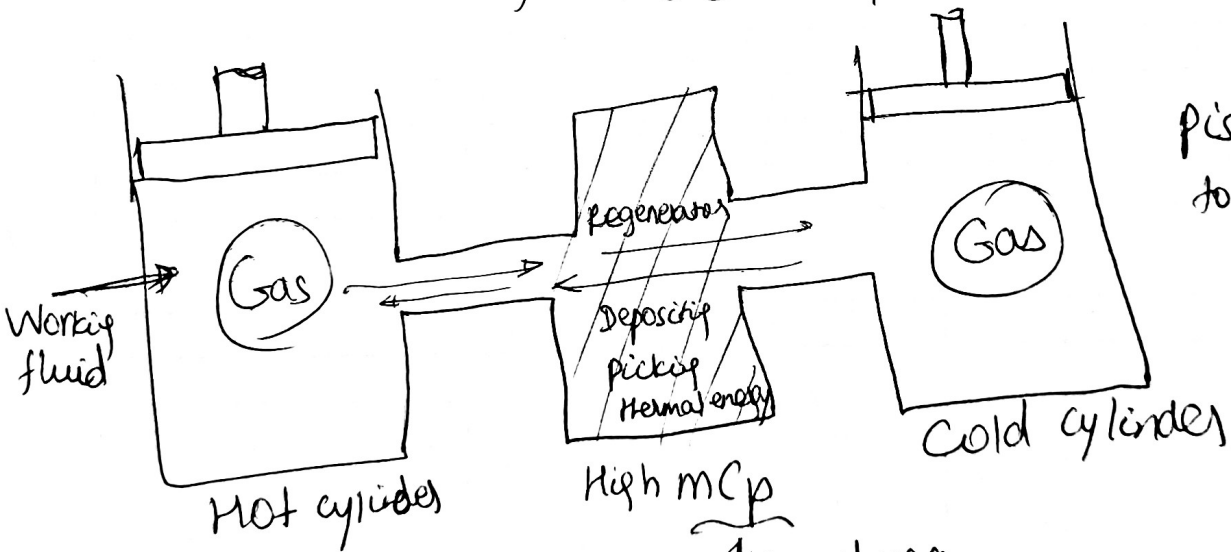


Pressure is going high:
 compression $\rightarrow P \uparrow, T \uparrow$
 Rejecting heat
 $\hookrightarrow$ will maintain T

Regenerator

Takes thermal energy from one to another.

$\hookrightarrow$ from 2-3 to 4 ~~to~~ 1



Working fluid - Helium
 Hydrogen
 $\uparrow$ Thermal mass